Report 4: Pulse Shapes, the Nyquist Condition and the End-to-End Operator

EE451 — Zak Transform / Delay–Doppler Communication
Simulation S4

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October 12, 2026

1 What this report is about

- In Report 3 the loopback was exact for a rectangular pulse. The course slides use smoother pulses (Gaussian × sinc, "orthogonalised" Gaussian).
- Do these pulses also give back exactly what was sent? If not, how large is the error and why?
- This report gives a simple test (the *Nyquist condition*) that predicts the error before simulating, and checks the prediction.

2 The concept in brief

- **Each symbol is a pulse "brick".** The transmitter places one copy of the pulse g per symbol. The receiver recovers a symbol by comparing (correlating) the received waveform with the same pulse.
- **The question is: does a neighbour leak into my measurement?** If the pulse of the symbol one delay bin away has a non-zero overlap with my pulse, it adds an error to my measurement.
- **Nyquist condition.** Let $R[k]$ be the overlap (autocorrelation) of the pulse with a copy of itself shifted by k delay bins (kT_s). Zero leakage needs

$$R[0] = 1, \qquad R[k] = 0 \ \text{ for } k \neq 0.$$

- **The end-to-end operator H.** With no channel, $Y = H\,X$. H is built only from the numbers $R[k]$. The condition above means $H = I$ (identity): every symbol comes back unchanged.
- **Size of the error.** $\varepsilon = \sqrt{\sum_k \left|R[k] - \delta_k\right|^2}$ is the rms error relative to the rms symbol size. In particular, $100\,\varepsilon\,\%$ is the expected EVM (error vector magnitude).
- **Leakage is only along delay.** A symbol can leak into other delay bins, but never into other Doppler bins (Doppler is a pure phase rotation across periods).
- **Normalisation.** The pulse must have unit energy, $\sum |g|^2 = 1$. Otherwise $Y \approx cX$ with $c = \sum |g|^2 \neq 1$.

Pulse	Idea
Rectangular (width T_s)	neighbouring pulses never overlap ⇒ perfect, but a poor spectrum (Report 6)
Gaussian × sinc ($\alpha = 4, 16$)	slides' pulse; smooth and band-limited, but its copies overlap
Orthogonalised Gaussian×sinc	the same pulse, slightly reshaped (in the frequency domain) so that its copies stop overlapping

One-line memory aid: the autocorrelation at multiples of T_s must be $1, 0, 0, \ldots$ (Nyquist). If so, $H = I$. If not, the rms error is $\varepsilon = \sqrt{\sum_k |R[k] - \delta_k|^2}$, and it only spreads along delay.

3 What was simulated

- Four pulses: rectangular, Gauss×sinc with $\alpha = 4$ and $\alpha = 16$, and the orthogonalised pulse ($\alpha = 16$), on a M=16, N=8 grid with D=4. (Only the constellation test of Fig. 4 uses a larger grid, M=32, N=16.)
- For each pulse: the autocorrelation test, the full operator H, and the loopback of random 16QAM symbols.
- The prediction ε from $R[k]$ against the measured error for many pulse lengths.

4 Results and discussion

4.1 The pulses

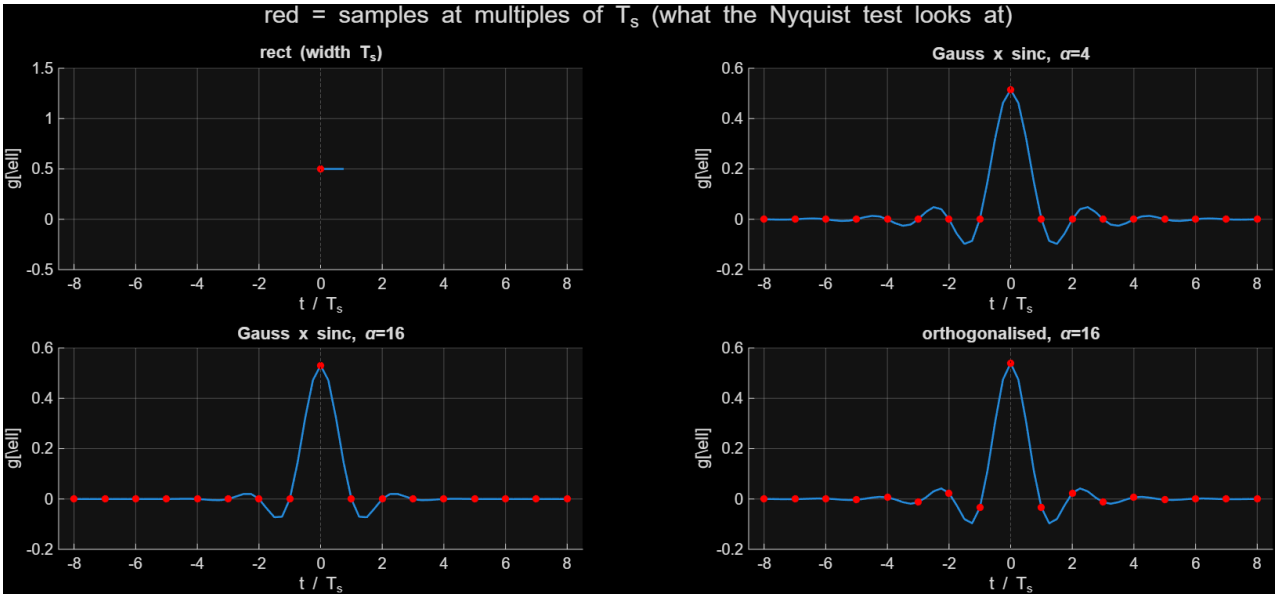


Figure 1: The four pulses. Red dots: pulse values at multiples of T_s.

- The rectangular pulse is a flat piece of width T_s (height $1/\sqrt{D} = 0.5$).
- The Gauss×sinc pulses are zero at all integer multiples of T_s except the centre (like a sinc). One might think this is enough for orthogonality; it is not. What counts is the autocorrelation, not the pulse values.
- The orthogonalised pulse is slightly different: its red dots near ± 1 and ± 2 are not exactly zero. That small adjustment is what makes its copies orthogonal.

4.2 The Nyquist test

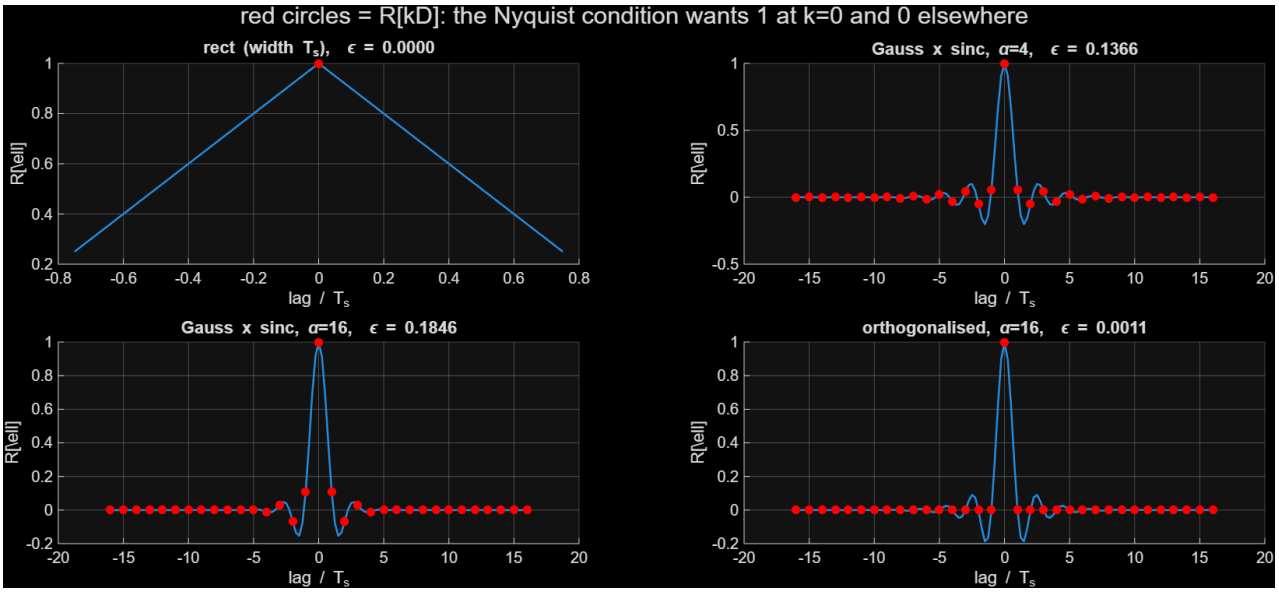


Figure 2: Autocorrelation of each pulse (blue) and its values at multiples of T_s (red). Nyquist wants 1 at k=0 and 0 elsewhere.

- **Expected:** rect and orthogonalised pass the test; Gauss×sinc fails it.
- **Observed:** $\varepsilon = 0$ (rect), 0.0011 (orthogonalised), 0.137 (α=4) and 0.185 (α=16). The red dots of the two Gauss×sinc pulses are visibly non-zero near the centre, while those of the orthogonalised pulse are essentially zero.
- Takeaway: smoothness alone does not give orthogonality; a plain Gauss×sinc has a built-in error of about 14–18%.

4.3 The end-to-end operator H

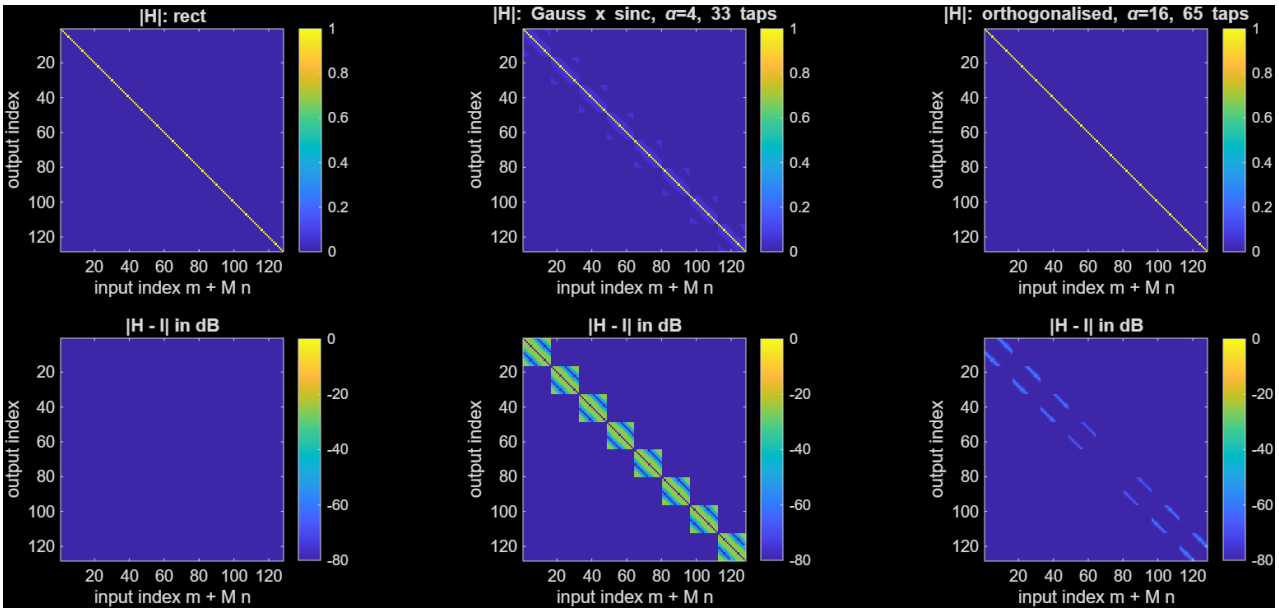


Figure 3: $|H|$ (top) and $|H - I|$ in dB (bottom) for M=16, N=8 (a 128 × 128 matrix). Left: rect. Middle: Gauss×sinc, α=4, 33 taps. Right: orthogonalised, 65 taps.

- **Expected:** identity for rect and (nearly) for the orthogonalised pulse; a smeared diagonal for Gauss×sinc.
- **Observed:** rect gives a perfect diagonal ($|H - I|$ is below the -80 dB floor). Gauss×sinc smears around the diagonal, only inside blocks of 16×16 (one block per Doppler bin), with the strongest elements at about -25 dB (the largest off-diagonal value is 0.057). The orthogonalised pulse is below -60 dB everywhere (7×10^{-4} at most).
- Takeaway: the blocks confirm that mixing is only along delay.

4.4 What the receiver sees

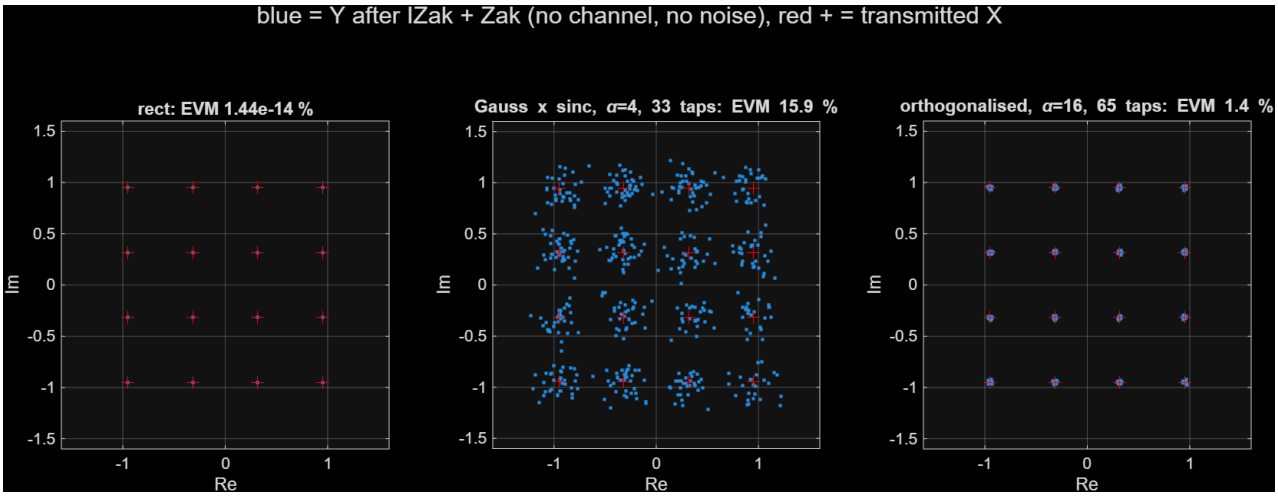


Figure 4: Recovered symbols (blue) against the transmitted 16QAM points (red), no channel, no noise. This test uses M=32, N=16.

- **Expected:** EVM $\approx 100\,\varepsilon\,\%$.
- **Observed:** rect: $10^{-14}\,\%$; Gauss×sinc (33 taps): 15.9 % against a prediction of 16.3 %; orthogonalised: 1.4 % against 1.5 %. These predictions differ from the M=16 values quoted elsewhere (0.157 and 0.0011) because the Gaussian width scales with $T = MT_s$, so ε depends on M.
- Takeaway: even without any channel or noise, a non-orthogonal pulse blurs the constellation into clouds; an orthogonalised pulse keeps it tight.

4.5 Leakage goes along delay only

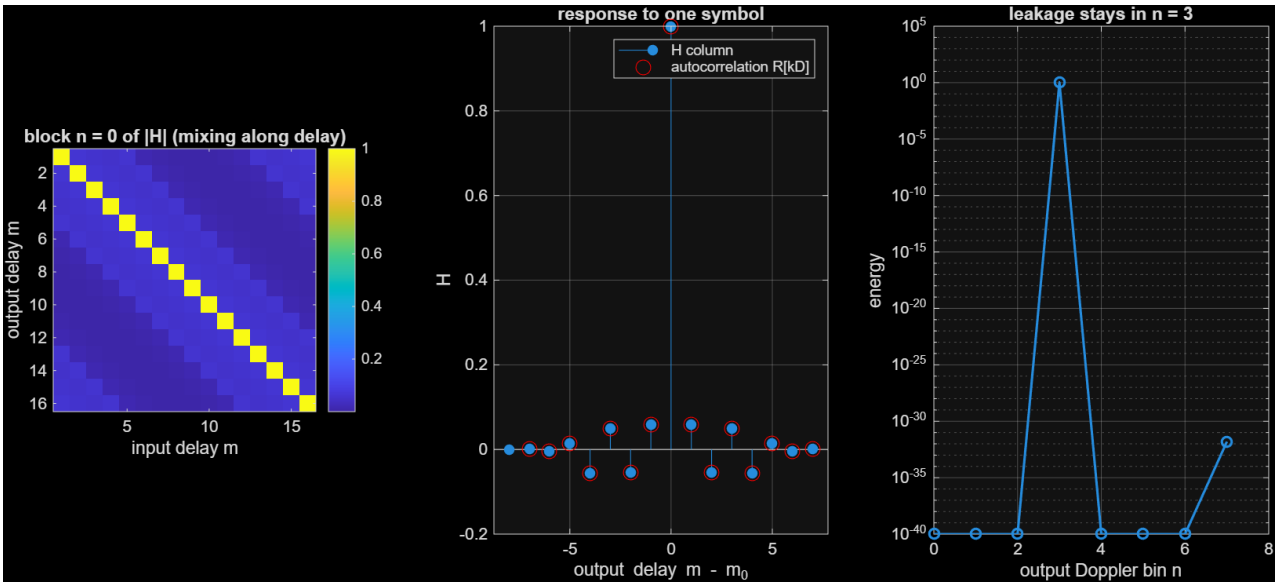


Figure 5: Left: one Doppler block of $|H|$. Middle: response to a single symbol (blue) against the autocorrelation values $R[k]$ (red). Right: energy per output Doppler bin.

- **Expected:** the response to one symbol equals $R[k]$ along delay and stays inside the symbol's own Doppler bin.
- **Observed:** the blue stems sit on the red circles, so the autocorrelation *is* the leakage profile. The energy in the symbol's own Doppler bin (n=3) is $1.0245 = 1 + \varepsilon^2$; all other Doppler bins are at 10^{-30} or below.
- Takeaway: this is why the single number ε describes the whole system.

4.6 Prediction against measurement

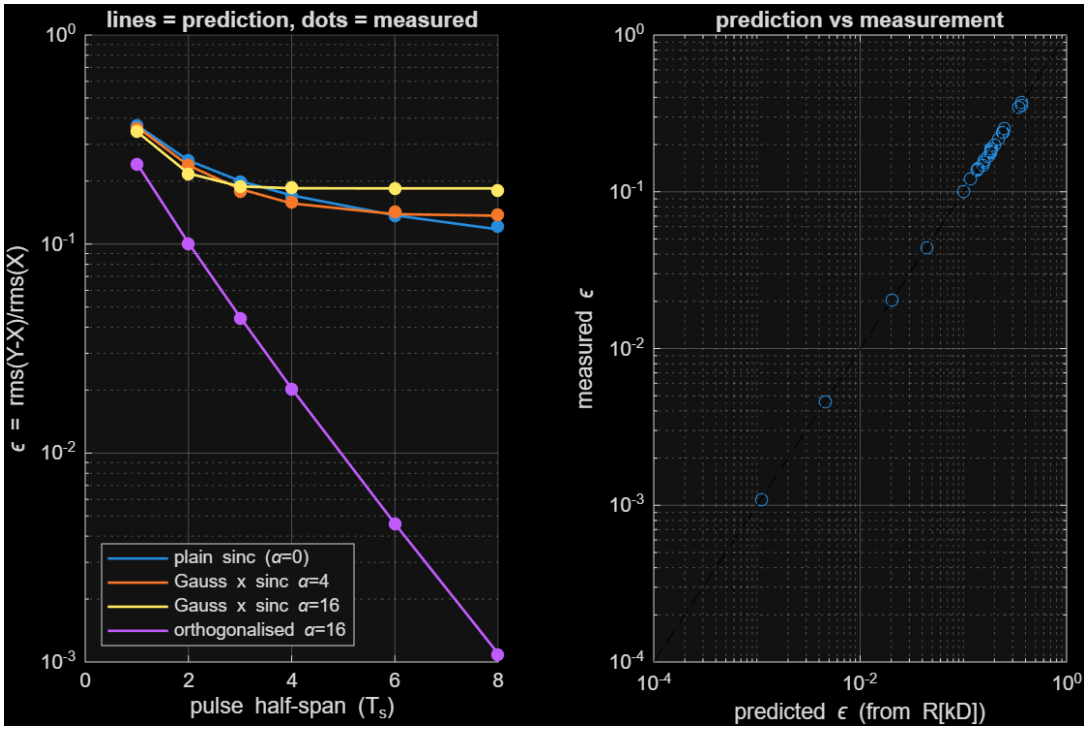


Figure 6: Left: ε against pulse half-span (lines: predicted from $R[k]$; dots: measured). Right: predicted against measured for all cases (diagonal = perfect agreement).

- **Expected:** the measured error follows the prediction from $R[k]$.
- **Observed:** the largest deviation is 3.3 % (the measurement uses a finite number of random frames). The matrix test is even exact: $\|H - I\|_F/\sqrt{MN}$ equals the predicted ε to all printed digits (0.1565 and 0.0011).
- The Gaussian-windowed pulses level off at about 0.14 (α=4) and 0.18 (α=16), so making them longer does not help; the plain sinc only improves slowly (0.12 at a half-span of $8\,T_s$). The orthogonalised pulse keeps improving, to $\varepsilon \approx 10^{-3}$ at a half-span of $8\,T_s$.
- Takeaway: the Nyquist test predicts the system's accuracy, and only an orthogonalised pulse approaches the ideal.

Table 1: Summary of the four pulses (D=4). The EVM test of Fig. 4 uses a larger grid (M=32, N=16), so its prediction differs from the M=16 one.

Pulse	ε (M=16)	EVM test (M=32): pred. / meas.
Rectangular	0	0 / ≈ 0
Gauss×sinc, α=4	0.137 (long), 0.157 (33 taps)	16.3 % / 15.9 % (33 taps)
Gauss×sinc, α=16	0.185	—
Orthogonalised, α=16, 65 taps	0.0011	1.5 % / 1.4 %

5 Summary

- The test is simple: the pulse autocorrelation at multiples of T_s must be $1, 0, 0, \ldots$; then $H = I$.
- Rectangular pulse: passes. Gauss×sinc (slides' pulse): fails, with $\varepsilon \approx 0.14$–0.18 that does not shrink with length. Orthogonalised version: passes ($\varepsilon \approx 10^{-3}$).
- The prediction and the simulation agree within 3.3 %; leakage is along delay only.

Appendix: details and assumptions

- The slides do not specify the pulse widths in Gauss×sinc. We assumed the sinc zero spacing equals T_s and the Gaussian width scales with $T = MT_s$. A different choice changes the numbers somewhat but not the conclusion.
- The slides only name the "orthogonalised" Gaussian; the recipe used here (flatten the folded power spectrum, then truncate) is our own construction, not given in the slides.
- Normalisation: for an un-normalised sinc-type pulse, $c = \sum |g|^2 \approx 3.77$ and the measured loop-back gain was within about 1.5 % of c, as expected.
- The EVM test of Fig. 4 uses its own, larger grid (M=32, N=16) with the pulse built for that M, so its predicted values (16.3 % and 1.5 %) differ from the M=16 matrix-test values (0.157 and 0.0011). With M=32 we reproduce 0.1631 exactly, and the measured 15.9 % and 1.4 % agree with the predictions.
- At $D = 1$ the sampled sinc collapses to a single sample and the test gives a misleadingly perfect result ($\varepsilon \approx 10^{-16}$); this trap is explained in Report 5.
- The figures come from `S4_pulses.m` (MATLAB dark theme). Fig. 6 shows the left and right panels of a three-panel figure; the middle panel (the effect of D) is omitted here and is the subject of Report 5.